Cell Injury Mechanisms

Biochemical mechanisms of cell injury

1. ATP depletion
2. Mitochondrial damage
3. Ca^{2+} influx and loss of calcium homeostasis
4. Accumulation of oxygen-derived free radicals (oxidative stress)
5. Defects in membrane permeability
6. Damage to DNA and proteins (misfolded proteins)

Phase 1: The immediate metabolic crisis (ischemia/hypoxia)

When blood flow is obstructed (ischemia) or oxygen supply drops (hypoxia), the cell's energy production collapses.

- **Failure of oxidative phosphorylation:** Mitochondria lack oxygen, the final electron acceptor of the electron transport chain. Electron flow halts, the proton gradient across the inner mitochondrial membrane collapses, and ATP synthesis falls sharply.
- **Acceleration of anaerobic glycolysis:** The cell switches to anaerobic glycolysis to make ATP from glycogen. This is highly inefficient and rapidly depletes glycogen stores.
- **Intracellular acidosis:** Anaerobic glycolysis causes accumulation of lactic acid and inorganic phosphates. Intracellular pH falls from about 7.2 to a more acidic range.
- **Chromatin clumping:** Low intracellular pH causes clumping of nuclear chromatin. (One proposed explanation: excess H^+ protonates basic residues on histones, altering their charge and the electrostatic interactions between chromatin fibers. For exams, "low pH → chromatin clumping" is the standard statement.)

Phase 2: Reversible injury, pump failure and hydropic change

As ATP falls below critical levels, energy-dependent membrane transport fails and the cell loses its ionic and osmotic balance. Membranes and organelles are swollen and stressed but remain physically intact.

Normal calcium handling. Cytosolic Ca^{2+} is kept very low (about 0.1 μM) compared with extracellular Ca^{2+} (about 1 to 2 mM). Two systems maintain this:

- **Sequestration into stores:** SERCA pumps Ca^{2+} from the cytosol into the ER using ATP. Mitochondria also take up Ca^{2+} through the mitochondrial calcium uniporter (MCU), driven by the negative inner membrane potential, and act as a buffer when cytosolic Ca^{2+} rises.
- **Extrusion from the cell:** PMCA pumps Ca^{2+} out using ATP. NCX removes Ca^{2+} using the inward Na^+ gradient (3 Na^+ in, 1 Ca^{2+} out). It is not directly ATP-powered, but it depends on the Na^+ gradient that the Na^+/K^+-ATPase maintains.

What fails in ischemia:

- **Failure of the Na^+/K^+-ATPase pump:** Without ATP, the pump can no longer extrude 3 Na^+ and import 2 K^+ per cycle. Na^+ enters and accumulates in the cell, while K^+ leaks out down its gradient.
- **Reversal of the Na^+/Ca^{2+} exchanger (NCX):** Normally NCX uses the inward Na^+ gradient to extrude Ca^{2+}. With the cell full of Na^+, the gradient flattens or reverses and NCX can run in reverse, expelling Na^+ but bringing Ca^{2+} into the cytosol.
- **Failure of primary calcium pumps:** The SERCA (ER) and PMCA (plasma membrane) Ca^{2+}-ATPases fail from lack of ATP, so calcium cannot be sequestered into the ER or pumped out.
- **Calcium is already rising in this phase.** Besides influx, Ca^{2+} is released from the ER and from damaged mitochondria into the cytosol. This early rise in cytosolic Ca^{2+} is a key driver of the transition to irreversible injury.

Why Ca^{2+} is released from the ER and mitochondria

Both organelles hold Ca^{2+} against a steep gradient and need energy and intact membranes to do so. Injury removes both.

- **ER:** SERCA normally pumps Ca^{2+} in against a constant leak back out. With ATP depleted, SERCA fails, so the leak is no longer offset and Ca^{2+} drains into the cytosol. Rising cytosolic Ca^{2+} and oxidative stress also sensitize the ER release channels (IP_3 and ryanodine receptors), so more Ca^{2+} is released (calcium-induced calcium release).
- **Mitochondria:** Early in injury they absorb Ca^{2+} through the MCU and buffer the rise. Hypoxia collapses the membrane potential, removing the driving force and ATP needed to retain calcium. Matrix Ca^{2+} overload, ROS and low ATP open the mitochondrial permeability transition pore (MPTP), so stored Ca^{2+} leaks into the cytosol and the mitochondria swell, worsening ATP failure.
- **Vicious cycle:** More cytosolic Ca^{2+} activates phospholipases, proteases and endonucleases, damages membranes and further impairs ATP production, which drives more Ca^{2+} release.

Histopathological and ultrastructural model: cellular swelling (hydropic change)

Trapped intracellular Na^+ raises osmotic pressure, and water enters the cell through aquaporins. The water distributes into three compartments:

1. **Cytosol (cloudy swelling):** Water expands the cytosol and dilutes the cytoplasm. The whole cell looks enlarged, and under the light microscope the cytoplasm looks pale and "cloudy."
2. **Endoplasmic reticulum (vacuolar degeneration):** The ER cisternae fill with fluid and dilate. Under the light microscope these appear as clear vacuoles in pale cytoplasm. Under TEM the ER is markedly dilated and ribosomes detach from the rough ER and disperse in the cytosol.
3. **Mitochondria (mitochondrial swelling):** Water enters the matrix and the mitochondria enlarge. Under TEM, small dark amorphous (protein-rich) densities may appear in the matrix, but the inner and outer membranes remain continuous.

Separate staining change: mild cytoplasmic eosinophilia. Ribosome detachment and early degradation of cytoplasmic RNA reduce binding of the basic dye hematoxylin (RNA is acidic and normally stains blue-purple). The cytoplasm therefore looks mildly more pink. This reflects relative loss of RNA, before major protein denaturation.

Additional reversible ultrastructural features and inclusions

- **Plasma membrane changes:**

    - **Loss of microvilli:** Surface extensions flatten as the membrane stretches.
    - **Membrane blebbing:** The membrane detaches locally from the weakened cytoskeleton and forms outward bulges (blebs).
    - **Loosening of intercellular junctions:** Cell-to-cell attachments begin to loosen.
- **Fatty change (steatosis), especially in liver and myocardium:** ATP depletion disrupts fatty acid oxidation and lipoprotein synthesis. Triglycerides accumulate as clear lipid vacuoles in the cytoplasm.
- **Contained myelin figures:** Whorled phospholipid masses derived from damaged membranes (resembling onion layers on TEM) form but remain inside the cell.

Phase 3: The irreversible turning point (pathway leading to necrosis)

If ischemia persists, the physical strain of swelling and the rising intracellular calcium push the cell past the point of no return. The hallmark is the **inability to reverse mitochondrial dysfunction** and **profound membrane damage**. Three groups of damage occur together:

The two points of no return

- **Irreversible mitochondrial dysfunction:** Ca^{2+} overload in the mitochondrial matrix (after uptake through the MCU, fed by the high cytosolic Ca^{2+}), together with ROS and low ATP, opens the MPTP. The membrane potential is lost and ATP production cannot be restored even if oxygen returns.
- **Profound membrane damage (plasma and lysosomal membranes):** The main causes are Ca^{2+}-activated phospholipases (which break down membrane phospholipids), ROS (lipid peroxidation) and osmotic swelling. Calpain-mediated cytoskeletal damage removes the membrane's support, and detergent-like phospholipid breakdown products (free fatty acids, lysophospholipids) add to the injury. The plasma membrane damage allows massive Ca^{2+} influx. Lysosomal rupture releases acid hydrolases, which then drive autolysis (digesting proteins, RNA and DNA). They amplify the damage and the necrotic changes rather than being the main cause of plasma membrane damage.

Sources of ROS (oxidative stress)

ROS drive much of the damage below. In ischemia and reperfusion they come mainly from damaged mitochondria, with additional enzymatic sources.

- **Mitochondria (main source):** A small fraction of electrons normally leaks from the electron transport chain (mainly complexes I and III) and reduces O_2 to superoxide ($O_2^{\bullet -}$), which antioxidant enzymes handle. In hypoxia the chain is damaged and backed up. When oxygen returns, electron leakage increases sharply and a burst of superoxide is produced. Ca^{2+} overload and MPTP opening worsen the damage and ROS production.
- **NADPH oxidase:** Neutrophils and macrophages produce superoxide in the respiratory burst. This adds ROS when neutrophils infiltrate the tissue, especially at reperfusion.
- **Xanthine oxidase:** During ischemia, ATP breakdown causes hypoxanthine and xanthine to accumulate, and xanthine dehydrogenase is converted to xanthine oxidase. On reperfusion, oxygen lets it make superoxide and H_2O_2.
- **Peroxisomal oxidases and cytochrome P450 enzymes:** These also generate H_2O_2 and superoxide.

- **Conversion to more reactive species:** Superoxide dismutase (SOD) converts superoxide to H_2O_2. The Fenton reaction ($Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + \bullet OH + OH^-$) converts H_2O_2 to the highly reactive hydroxyl radical (•OH).
- **Other causes of ROS (not specific to ischemia):** radiation, chemicals and toxins (for example CCl_4), and inflammation.

1. Membrane rupture and structural dissolution

- **Plasma membrane damage:** Cell swelling plus activation of calcium-dependent phospholipases breaks down membrane phospholipids. The membrane develops physical defects and loses integrity.
- **Massive calcium influx:** Extracellular Ca^{2+} (about 1 to 2 mmol/L) is roughly 10,000-fold higher than cytosolic Ca^{2+} (about 0.1 μmol/L). Once the plasma membrane is breached, calcium floods in down this gradient. Note that cytosolic calcium was already elevated in Phase 2. Membrane rupture makes the rise massive.
- **Lysosomal membrane rupture:** Swelling and free radical damage break lysosomal membranes.
- **Spillage of acid hydrolases:** Proteases, nucleases, cathepsins and phosphatases leak into the cytosol. The acidic cytosol created in Phase 1 favors their activity, so they begin autolysis (self-digestion).
- **Irreversible mitochondrial damage:** On TEM, large dense flocculent densities appear in the matrix, with disruption of cristae and rupture of mitochondrial membranes.
- **Uncontained myelin figures:** Myelin figures become abundant and escape the cell.

2. Protein modification and cross-linking

- **Protease activation:** The calcium surge activates calpains and other proteases that degrade cytoskeletal proteins (for example actin, tubulin, vinculin), detaching the cytoskeleton from the plasma membrane.
- **Oxidative modification:** Reactive oxygen species (ROS) attack amino acid side chains. Cysteine sulfhydryl (-SH) groups are oxidized and can form abnormal disulfide bonds. Methionine residues are oxidized to methionine sulfoxide.
- **Aggregation:** Abnormal disulfide cross-links and misfolding produce insoluble protein aggregates and loss of enzyme function.

3. Severe DNA damage

- **Strand breaks:** Hydroxyl radicals (•OH) attack the sugar-phosphate backbone, causing single-strand and double-strand DNA breaks.
- **Base oxidation (8-OHdG):** ROS oxidize guanine to 8-hydroxy-2'-deoxyguanosine. During replication, 8-OHdG tends to mispair with adenine instead of cytosine, causing G:C to T:A mutations.

Phase 4: The branching point: necrosis vs. apoptosis

The route of cell death depends on the severity and nature of the injury, and on whether the plasma membrane is still intact and ATP is still available. **These are alternative pathways. Apoptosis requires an intact plasma membrane and some residual ATP, so it occurs instead of, not after, the plasma membrane rupture described in Phase 3.** A cell whose membrane has already been torn open dies by necrosis. Cells with severe DNA damage or misfolded proteins but a still-sealed membrane and preserved ATP can undergo apoptosis (for example, cells at the margin of an ischemic zone).

Path A: Severe, ATP-depleting injury: necrosis

With little or no ATP and a breached membrane, the cell dies by uncontrolled dissolution driven by the enzymes activated in Phase 3. On H&E staining:

- **Intense cytoplasmic eosinophilia:** The cell progresses from mild eosinophilia to a bright pink-red stain. Hematoxylin binding is lost because RNA is degraded, and eosin binding increases because denatured proteins expose more basic groups (lysine, arginine) that bind the acidic dye eosin.
- **Homogeneous, glassy cytoplasm:** Loss of organelles and cytoskeletal structure leaves a smooth, featureless pool of denatured protein.
- **Nuclear changes:**
    a. **Pyknosis:** The nucleus shrinks and becomes a dense, dark mass of condensed chromatin.
    b. **Karyorrhexis:** The pyknotic nucleus fragments.
    c. **Karyolysis:** Nucleases digest the chromatin and the nucleus fades and disappears.
- **Lysis and inflammation:** The cell breaks down and its contents spill into the extracellular space. Released material acts as damage-associated molecular patterns (DAMPs) that trigger a local inflammatory response.

Path B: Preserved membrane integrity and residual ATP: intrinsic (mitochondrial) apoptosis

1. **p53 activation:** DNA damage stabilizes and activates p53. p53 first causes cell-cycle arrest and promotes DNA repair. If the damage cannot be repaired, p53 promotes apoptosis by upregulating pro-apoptotic genes such as Bax, PUMA and Noxa (Bak is also involved).
2. **Mitochondrial outer membrane permeabilization (MOMP):** Bax and Bak oligomerize and form channels in the outer mitochondrial membrane.
3. **Release of pro-apoptotic factors:** Cytochrome c and Smac/DIABLO leak from the intermembrane space into the cytosol.
4. **Removal of the brakes:** Smac/DIABLO binds and neutralizes IAPs (inhibitors of apoptosis proteins), releasing inhibition of caspases.
5. **Apoptosome assembly:** In the presence of dATP/ATP, cytochrome c binds Apaf-1. Seven Apaf-1 complexes assemble into a wheel-shaped apoptosome.
6. **Caspase cascade:** Procaspase-9 is recruited to the apoptosome and activated by induced proximity and dimerization. Active caspase-9 (the initiator) then cleaves and activates the executioner caspases, caspase-3 and caspase-6/7.
7. **Orderly dismantling:** Caspase-3 cleaves the inhibitor of caspase-activated DNase (CAD/ICAD), so CAD enters the nucleus and cuts DNA into regular fragments (internucleosomal cleavage). Executioner caspases also cleave the cytoskeleton, and the cell breaks into membrane-bound apoptotic bodies.
8. **Phagocytic clearance:** Phosphatidylserine flips to the outer membrane leaflet, acting as an "eat-me" signal. Macrophages engulf the apoptotic bodies. Because cell contents do not leak out, there is no inflammation.

Phase 5: The paradox of reperfusion injury

Restoring oxygenated blood flow (reperfusion) to cells that are damaged but still viable can paradoxically cause additional injury, and can convert reversible injury into necrosis in some cells.

- **Calcium overload:** Reperfusion delivers calcium-rich blood to cells with already leaky membranes and impaired calcium handling, causing further calcium overload.
- **Opening of the mitochondrial permeability transition pore (MPTP):** Calcium overload and oxidative stress open the MPTP, a large channel in the inner mitochondrial membrane. The mitochondrial membrane potential is lost, ATP production fails, and the mitochondria swell and release pro-death factors such as cytochrome c. Depending on remaining ATP, this leads to necrosis or apoptosis.
- **Burst of ROS:** Oxygen reaching damaged, uncoupled electron transport chains is incompletely reduced, generating superoxide ($O_2\bullet^-$), hydrogen peroxide (H_2O_2) and hydroxyl radicals (•OH). Xanthine oxidase (formed from xanthine dehydrogenase during ischemia, with hypoxanthine accumulating from ATP breakdown) adds more superoxide and H_2O_2 once oxygen returns. This accelerates lipid peroxidation, protein cross-linking and DNA damage (including 8-OHdG), and can push damaged cells toward necrosis.
- **Inflammation and neutrophil amplification:** Dying cells release cytokines and DAMPs. With blood flow restored, neutrophils infiltrate the tissue and, through respiratory burst (NADPH oxidase) and release of lysosomal enzymes, produce more ROS and damage, including to surrounding viable tissue.

Summary of changes from the original

1. **Na^+/K^+ pump:** corrected the garbled sentence. Na^+ accumulates and K^+ leaks out.
2. **Swelling compartments:** now three compartments (cytosol, ER, mitochondria). Eosinophilia is listed as a separate staining change.
3. **8-OHdG:** causes G:C to T:A mutations through mispairing, not "cross-linking."
4. **Cysteine and methionine:** only cysteine forms disulfide bonds. Methionine is oxidized to a sulfoxide.
5. **p53:** causes cell-cycle arrest and repair first, then apoptosis. It does not "halt all metabolic activity." Bax, PUMA and Noxa are mentioned.
6. **Phase 4:** apoptosis is now explicitly an alternative to membrane rupture, not a sequel to it.
7. **Calcium timing:** cytosolic Ca^{2+} already rises in the reversible phase. Membrane rupture makes the influx massive.
8. **Reperfusion:** wording softened from "forcing viable cells" to "can convert," and the MPTP description no longer claims permanent, absolute loss of ATP production.
9. **Chromatin clumping and caspase-9:** added the standard exam wording and the correct activation mechanism (induced proximity and dimerization).